

DSAgent

Autonomous Pipeline Report

Dataset: titanic.csv

Generated: May 10, 2026 at 12:15 PM

Session: 9f11ec80-8b4...

7

Steps Executed

6

Phases Completed

17.5s

Total Time

1. Dataset Overview

The dataset contains **891** rows and **12** columns, using **0.28 MB** of memory.

Type	Count	Columns
Numeric	7	PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
Categorical	5	Name, Sex, Ticket, Cabin, Embarked

Numeric Statistics

Column	Mean	Median	Std	Min	Max
PassengerId	446.00	446.00	257.35	1.00	891.00
Survived	0.38	0.00	0.49	0.00	1.00
Pclass	2.31	3.00	0.84	1.00	3.00
Age	29.70	28.00	14.53	0.42	80.00
SibSp	0.52	0.00	1.10	0.00	8.00
Parch	0.38	0.00	0.81	0.00	6.00
Fare	32.20	14.45	49.69	0.00	512.33

AI Analysis

LLM error: HF_API_KEY not configured

2. Data Quality Assessment

Found **3** columns with missing values out of 891 rows.

Column	Missing Count	Missing %
Age	177	19.87%
Cabin	687	77.1%
Embarked	2	0.22%

Potential Issues:

- High cardinality categorical columns: ['Name']

6. Model Training & Comparison

Problem Type: **regression** | Target: **Fare** | Features: **11**

Best Model: **Linear Regression** — Score: **46.8%**

Model	R ²	RMSE	MAE
Random Forest	0.1887	35.4328	13.4882
XGBoost	-0.1671	42.4976	14.1759
LightGBM	0.3771	31.0471	13.1623
Linear Regression	0.4675	28.7044	18.3167

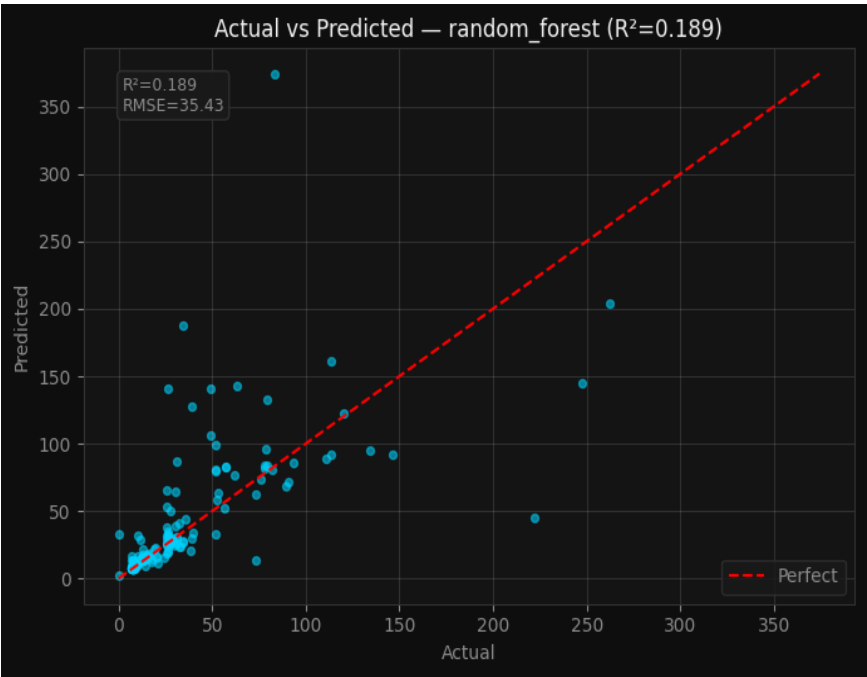
Model Selection Rationale

LLM error: HF_API_KEY not configured

7. Model Evaluation

Model Evaluation Metrics

r2_score: 18.9% | rmse: 35.4328 | mae: 13.4882



Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

LLM error: HF_API_KEY not configured

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